

Supporting Information:

Catalytic competence bounds the decay rates of a thermally formed radical pair

Hikaru Wakaura and Taiki Tanimae

S1 Software and environment

Spin dynamics: `qbscreen` (this work) with NumPy and SciPy; coherent yields from the Sylvester equation $H_{\text{eff}}X - XH_{\text{eff}}^\dagger = -i\rho_0$ with $H_{\text{eff}} = 2\pi H - \frac{i}{2}(k_S P_S + k_P)$ and $X = \int_0^\infty \rho dt$; yields with electron spin relaxation from the Liouville-space generator. The two agree to 10^{-8} without relaxation, and every yield pair sums to one. Electronic structure: PySCF 2.12.1 (B3LYP; def2-SVP for couplings and energies, def2-TZVP for hyperfine couplings; RI-J; PCM), geometries from GFN2-xTB 6.7.1. Every table below is generated from the stored results by `qbscreen/make_si_v6.py`.

S2 The inequality $\Phi_P \leq 4k_P/k_S$ and the dead-end bound

Let $\mathcal{L}$ be the generator of eq 2 of the main text, with $\mathcal{D}$ generating a positivity-preserving evolution (for example a trace-preserving GKSL generator) and unital, $\mathcal{D}(I) = 0$. Then $-\mathcal{L}(I) = k_S P_S + k_P I$. For $k_P > 0$ the operator $\mathcal{R} = (-\mathcal{L})^{-1} = \int_0^\infty e^{\mathcal{L}t} dt$ exists and is positive, because $e^{\mathcal{L}t}$ is and decays at least as $e^{-k_P t}$. Applying $\mathcal{R}$ to the identity above gives

$$k_S \mathcal{R}(P_S) = I - k_P \mathcal{R}(I) \preceq I, \quad (\text{S1})$$

so $\text{Tr } \mathcal{R}(P_S) \leq 4N/k_S$ with $4N$ the dimension of the spin space, and $\Phi_P = (k_P/N) \text{Tr } \mathcal{R}(P_S) \leq 4k_P/k_S$.

For a dead-end pair ($k_P = 0$) the resolvent need not exist: without nuclei and with $H = 0$, for example, the 16×16 generator has rank 7, because triplet states that never reach the singlet are stationary. Integrating $\frac{d}{du} e^{\mathcal{L}u}(I) = -k_S e^{\mathcal{L}u}(P_S)$ instead gives

$$k_S \int_0^t e^{\mathcal{L}u}(P_S) du = I - e^{\mathcal{L}t}(I) \preceq I \quad (\text{S2})$$

for all t , since $e^{\mathcal{L}t}(I) \succeq 0$. The population built up from a constant singlet source $k_f[\mathbf{E} \cdot \mathbf{S}]P_S/N$ is therefore at most $4k_f[\mathbf{E} \cdot \mathbf{S}]/k_S = e^{-\Delta/k_B T}[\mathbf{E} \cdot \mathbf{S}]$ at all times.

`competence_sensitivity.random_bound_test` (seed 20261001; inputs and outputs in `calibration_results/phi_bound_test.json`) evaluates $\Phi_P k_S / 4k_P$ for 300 random spin systems: 0–3 nuclei of spin $\frac{1}{2}$ or 1 with couplings of 1–300 MHz, exchange 10^{-1} – 10^4 MHz of either sign, axial dipolar tensors of -10^{-1} to -10^3 MHz with random axes, random fields up to 0.1 T and rates, and, for about 30 % of the small systems, single-electron dephasing with T_2^e from 1 ns to 10 μ s. The largest value is 0.9986.

S3 The search for F

For each (k_S, k_P) on a grid of 15 by 13 rates, the exchange coupling is scanned on 99 points with sinh spacing (40 MHz scale), symmetric about zero and extending to $\max(4000 \text{ MHz}, 5(k_S + k_P)/2\pi)$, for dipolar scales $s = 0, 0.3, 1$ and 3 and 25 field directions (the pole and eight azimuths over 2π at polar angles of 30, 60 and 90°; the yield is invariant under $B \rightarrow -B$). Each of the three largest local maxima in J is refined twice on 21-point subgrids, and the four best distinct candidates are polished by a Nelder–Mead search in (J, θ, ϕ, s) with s clipped to $[0, 3]$ (160 evaluations each). A cell whose

maximum lies within 5 % of the range limit is searched again with a range of sixteen times the position of the maximum. Because the response has narrow resonances in the dipolar scale (at $k_S = 10^4$, $k_P = 10^{-2} \mu\text{s}^{-1}$ and $J \approx 0$ it is 0.40, 0.14, 0.68 and 0.04 % at $s = 0, 0.003, 0.01$ and 0.03 at the first-pass optimum; `calibration_results/s_resonance_example.json`), every cell is then searched a second time: s on 41 values (0 and 40 logarithmically spaced from 10^{-3} to 3), J on 21 points of the same sinh grid plus the first-pass optimum, all 25 directions, followed by a polish of the four best candidates (`worst_case_map.supplement`). The cell keeps the larger value. The second pass raised 151 cells, by up to a factor of 3.6, without changing any row maximum by more than 0.0 %. Repeating it with 121 values of s at the cells that set the thresholds changed F by a factor of at most 1.018 (`calibration_results/s_convergence.json`). Table S1 lists F in percent.

The bound G of the main text is computed from these grid values and is not a bound on the continuous rate region. At $\Delta = 0.46$ eV and $k_{\text{cat}} = 1 \text{ s}^{-1}$, k_{min} equals the grid rate $k_P = 31.6 \mu\text{s}^{-1}$; the permitted off-grid point $k_S = 76 \mu\text{s}^{-1}$, $J = 0$, $s = 0$, field along z , gives 0.0982 %, against $G = 0.0965$ % (`calibration_results/g_interval_example.json`).

Table S1: $F(k_S, k_P)$ (%). Rates in μs^{-1} .

$\log k_P \backslash \log k_S$	-2.0	-1.5	-1.0	-0.5	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
-2.0	6.3	12	14	13	10	7.5	6	5.2	4.6	3.7	2.3	1.3	1.3	1.3	1.3
-1.5	2.5	6.2	11	14	13	9.6	7.1	5.3	3.8	2.3	1.3	1.3	1.3	1.2	1.2
-1.0	0.85	2.4	6	11	13	12	8.4	5.4	2.7	1.1	1	1.1	1.1	1	1
-0.5	0.26	0.8	2.3	5.5	9.7	11	9	5.2	2	0.83	0.9	0.78	0.63	0.54	0.42
0.0	0.069	0.22	0.66	1.9	4.4	7.2	7.4	4.4	1.6	0.59	0.62	0.63	0.59	0.43	0.31
0.5	0.014	0.044	0.14	0.41	1.1	2.5	3.6	2.7	1.1	0.37	0.34	0.37	0.35	0.26	0.15
1.0	0.0014	0.0044	0.014	0.043	0.13	0.34	0.66	0.75	0.48	0.2	0.07	0.084	0.088	0.077	0.045
1.5	5.8e-5	1.8e-4	5.8e-4	0.0018	0.0056	0.017	0.043	0.083	0.096	0.062	0.032	0.049	0.055	0.042	0.02
2.0	1.7e-6	5.4e-6	1.7e-5	5.4e-5	1.7e-4	5.3e-4	0.0015	0.004	0.0077	0.014	0.018	0.02	0.023	0.015	0.0043
2.5	1.2e-7	3.8e-7	1.2e-6	3.8e-6	1.2e-5	3.7e-5	1.2e-4	3.5e-4	9.6e-4	0.0021	0.0033	0.0039	0.0036	0.0014	2.6e-4
3.0	4.8e-9	1.5e-8	4.8e-8	1.5e-7	4.8e-7	1.5e-6	4.8e-6	1.5e-5	4.4e-5	1.1e-4	2.3e-4	2.6e-4	1.4e-4	3.3e-5	4.5e-6
3.5	1.0e-10	3.2e-10	1.0e-9	3.2e-9	1.0e-8	3.2e-8	1.0e-7	3.1e-7	9.5e-7	2.6e-6	5.3e-6	5.4e-6	2.2e-6	3.0e-7	5.7e-8
4.0	1.1e-12	2.6e-12	7.6e-12	2.3e-11	7.4e-11	2.3e-10	7.4e-10	2.3e-9	7.2e-9	2.2e-8	5.8e-8	1.2e-7	1.1e-7	3.3e-8	3.2e-9

S4 Dependence on the spin model

For each onward rate that sets the thresholds of G , the cell that gives the row maximum of F in the reference model was searched again in richer models (`qbsscreen/model_sensitivity.py`). The search starts from the reference optimum and a 39-point J grid at its field direction and dipolar scale, and polishes the two best candidates. In “aniso” N5 and N10 carry the literature isotropic couplings plus the density-functional spin-dipolar tensors (Table S5), rotated into the frame of the contact complex in which the dipolar tensor was computed (0.09 Å root-mean-square deviation between the relaxed anion and the complex flavin). The complex was built with methylamine, so the benzylaminium tensors have no orientation in it: the aminium α -protons are kept isotropic (their anisotropy is below 5 % of the isotropic part), and in “+¹⁴N” the aminium nitrogen (spin 1), whose tensor is nearly axial about the singly occupied orbital, is added with its axis along the lone pair of the placed amine. “+2H” adds the two largest flavin methyl protons. The relaxation test uses a reduced pair (N5 isotropic, spin 1; aminium proton 256.9 MHz) searched with the full procedure, without relaxation and with single-electron dephasing (Table S2); the reduced-pair relaxation test is superseded by the direct four-nucleus calculation of section S5 and is shown for comparison. Only the reference row-maximum cell was searched again in each richer model, so another cell of the same row may give a larger value; the threshold shift quoted in the main text multiplies every searched value by the largest nuclear-model factor found and is a stress test, not a bound.

S5 Electron spin relaxation in the four-nucleus model

The four-nucleus spin space has dimension 144, too large for the dense Liouville-space solution. With $\gamma = 2/T_2^e$, single-electron dephasing $\gamma \sum_e (S_{ez} \rho S_{ez} - \rho/4)$ is split into a decay $-(\gamma/2)\rho$, absorbed into

Table S2: Searched F (%) at the threshold-setting cells in the reference and richer models, and in the reduced pair with and without electron dephasing.

$\log k_P$	$\log k_S$	reference	aniso	$+^{14}\text{N}$	+2H	red. coh.	10 ns	100 ns	1 μs	10 μs
1.0	1.5	0.75	2.3	1	1.6	0.51	0.21	0.22	0.41	0.5
1.5	2.0	0.096	0.28	0.16	0.2	0.086	0.052	0.038	0.076	0.085
2.0	4.0	0.023	0.0035	0.0033	0.0036	0.0033	0.016	0.0028	0.0032	0.0032
2.5	3.5	0.0039	0.0023	0.0015	0.0022	7.8e-4	0.0012	7.6e-4	7.8e-4	7.8e-4

$K = 2\pi H - \frac{i}{2}[k_S P_S + (k_P + \gamma/2)I]$, and a recycling term $\gamma \sum_e S_{ez} \rho S_{ez}$. The integral $X = \int_0^\infty \rho dt$ then solves $KX - XK^\dagger = -i\rho_0 - i\gamma \sum_e S_{ez} X S_{ez}$, which is iterated by GMRES with the left side inverted by a triangular Sylvester solve in the Schur basis of K (`product_yield.yields_t2e`). On twelve random systems small enough for the dense solution, including anisotropic hyperfine tensors, the two agree to a relative 1.3×10^{-13} . For every cell with $k_S \geq k_P$ on the threshold rows, the first-pass search of section S3 was repeated with $T_2^e = 10$ ns, 100 ns and 1 μs , with the same edge rule for the J range (Table S3; `calibration_results/worst_case_v6_t2e.json`). Nineteen searches (14 at $T_2^e = 10$ ns, 5 at 100 ns) first peaked at the edge of the J range and were repeated with the range widened sixteenfold; one of them ($k_S = 10^{4.5}$, $k_P = 10 \mu\text{s}^{-1}$, $T_2^e = 100$ ns) still peaks at the edge because the response approaches a plateau as $|J| \rightarrow \infty$: it is constant to five figures for $|J| = 10^6$ – 10^8 MHz and equal to the stored value (`calibration_results/t2e_plateau_check.json`).

Table S3: Searched F (%) without and with single-electron dephasing in the four-nucleus model. Rates in μs^{-1} .

$\log k_P$	$\log k_S$	coherent	10 ns	100 ns	1000 ns
1.0	1.0	0.66	0.075	0.22	0.54
1.0	1.5	0.75	0.12	0.29	0.63
1.0	2.0	0.48	0.18	0.26	0.41
1.0	2.5	0.2	0.21	0.18	0.18
1.0	3.0	0.07	0.22	0.21	0.067
1.0	3.5	0.084	0.22	0.21	0.073
1.0	4.0	0.088	0.22	0.21	0.076
1.0	4.5	0.077	0.22	0.21	0.065
1.0	5.0	0.045	0.22	0.21	0.04
1.5	1.5	0.083	0.024	0.046	0.077
1.5	2.0	0.096	0.042	0.059	0.09
1.5	2.5	0.062	0.063	0.045	0.059
1.5	3.0	0.032	0.072	0.028	0.03
1.5	3.5	0.049	0.075	0.035	0.047
1.5	4.0	0.055	0.076	0.04	0.053
1.5	4.5	0.042	0.077	0.033	0.041
1.5	5.0	0.02	0.077	0.015	0.019
2.0	2.0	0.0077	0.0067	0.0068	0.0076
2.0	2.5	0.014	0.011	0.01	0.014
2.0	3.0	0.018	0.013	0.014	0.017
2.0	3.5	0.02	0.015	0.016	0.019
2.0	4.0	0.023	0.015	0.018	0.022
2.0	4.5	0.015	0.015	0.012	0.014
2.0	5.0	0.0043	0.015	0.0033	0.0042
2.5	2.5	0.0021	0.0016	0.002	0.0021
2.5	3.0	0.0033	0.0024	0.0031	0.0032
2.5	3.5	0.0039	0.0023	0.0036	0.0039
2.5	4.0	0.0036	0.0017	0.0031	0.0035
2.5	4.5	0.0014	0.0013	0.0012	0.0013
2.5	5.0	2.6e-4	0.0012	2.2e-4	2.5e-4

S6 Hyperfine couplings

Fermi-contact couplings (B3LYP/def2-TZVP at GFN2-xTB geometries), all nuclei with $|a_{\text{iso}}| \geq 1$ MHz (Table S4), and the spin-dipolar tensors of the nitrogens and the largest protons (Table S5), computed from the spin density with the operator-derivative integrals $\langle \mu | \partial_a \partial_b | \mathbf{r} - \mathbf{R} |^{-1} | \nu \rangle$; their trace reproduces $-4\pi\rho_s(\mathbf{R})$. The computed flavin N5 and N10 isotropic couplings are within 16 % and 13 % of the literature values used in the main text,[1] and the N5 tensor is nearly axial with its large component normal to the ring. For 3-methylindole $\bullet^+$ the two large methyl couplings stand in for the tryptophan β -protons; their values depend on the side-chain conformation.

Table S4: Isotropic hyperfine couplings.

radical	nucleus (atom index)	a_{iso} (MHz)	a_{iso} (μT)
lumiflavin $\bullet^-$	H29	+17.9	+639
	H30	+17.8	+636
	N4	+12.4	+441
	H25	+11.3	+403
	H26	+11.3	+402
	H22	-10.5	-374
	H20	-5.3	-190
	H21	-5.3	-190
	N13	+4.6	+165
	H27	+1.4	+50
	N8	-1.0	-36
methylaminium $\bullet^+$	H2	+363.1	+12955
	H6	+82.7	+2951
	H5	+81.9	+2923
	H3	-59.3	-2116
	H4	-59.3	-2115
	N1	+30.5	+1090
benzylaminium $\bullet^+$	H10	+256.9	+9167
	H11	+196.2	+7001
	N0	+24.8	+885
	H8	-23.0	-821
	H9	-20.7	-737
	H14	-12.3	-439
	H16	-7.8	-278
	H13	-3.2	-115
	H15	+2.4	+85
3-methylindole $\bullet^+$	H12	+60.1	+2143
	H11	+55.5	+1980
	H18	-14.3	-511
	H13	-13.1	-467
	H14	-13.1	-466
	H16	-9.8	-351
	H15	-4.9	-175
	N3	+4.8	+172
	H17	+2.7	+95

S7 Dipolar tensor

The tensor is the point-dipole tensor summed over pairs of atoms weighted by their Löwdin spin populations (UKS B3LYP/def2-SVP on each radical at its position in the complex), for methylamine on the flavin si face with N5–N = 3.5 Å (Table S6).

Table S5: Spin-dipolar hyperfine tensors.

radical	nucleus	a_{iso} (MHz)	principal values of A_{dip} (MHz)
lumiflavin $\bullet^-$	H29	+17.9	-1.2, -1.0, +2.2
	H30	+17.8	-1.2, -1.0, +2.2
	N4	+12.4	-17.0, -16.7, +33.7
	H25	+11.3	-1.5, -0.9, +2.4
	H26	+11.3	-1.5, -0.9, +2.4
	N13	+4.6	-6.0, -5.7, +11.7
	N8	-1.0	-0.2, +0.1, +0.1
	N11	-0.2	-0.6, -0.4, +1.1
	H10	+256.9	-8.0, -4.2, +12.2
	H11	+196.2	-7.9, -4.2, +12.1
benzylaminium $\bullet^+$	N0	+24.8	-18.8, -18.6, +37.4
	H8	-23.0	-21.8, -6.0, +27.8
	H9	-20.7	-21.4, -5.4, +26.8

Table S6: Dipolar coupling of the contact pair.

model	D (MHz)	E (MHz)	principal values of T (MHz)
spin-population tensor	-843	-26	-1124, 511, 613
point dipole at spin centroids	-1642	0	–
point dipole at N5 and N	-1817	0	–

S8 Electronic coupling

Fragment-orbital couplings $t = \langle \text{LUMO}_{\text{Fl}} | F | \text{HOMO}_{\text{am}} \rangle$ (B3LYP/def2-SVP) for methylamine and benzylamine stacked on the flavin si face (five orientations) or approaching its edge (three orientations), at N5–N distances of 3.0–6.0 Å. Geometries with any intermolecular contact below 2.0 Å are excluded (Table S7). The edge approach of benzylamine clashes at every distance below 6 Å and is not used.

S9 Density-functional driving force

The main text uses measured potentials. For comparison, Table S8 gives adiabatic one-electron potentials in water ($\varepsilon = 78.4$, absolute SHE potential 4.44 V[2]) from the same density-functional protocol, against the measured values for a dialkylamine[2] and for free FMN at pH 7.[3] The flavin comparison is between different species (lumiflavin anion radical versus the pH-7 FMN oxidized/semiquinone couple) and indicates only the order of the error. Table S9 gives the raw driving force in low-dielectric continua with a contact Coulomb term $-e^2/(4\pi\varepsilon_0\varepsilon r)$. No zero-point or thermal corrections are included, and none of these values enters the main text.

S10 Cryptochrome

Orientation-averaged, perpendicular and parallel response of the cryptochrome pair over the exchange prefactor J_0 (both signs of J_{EH} ; $J = -2J_{\text{EH}}$, $J_{\text{EH}} = J_0 e^{-\beta r}$, $\beta = 14 \text{ nm}^{-1}$, $r = 1.90 \text{ nm}$ [4]), $k_S = k_P = 1 \mu\text{s}^{-1}$, coherent (Table S10).

References

- [1] A. A. Lee, J. C. S. Lau, H. J. Hogben, T. Biskup, D. R. Kattnig, and P. J. Hore. Alternative radical pairs for cryptochrome-based magnetoreception. *J. R. Soc. Interface.*, 11:20131063, 2014.
- [2] M. Jonsson, D. D. M. Wayner, and J. Lusztyk. Redox and acidity properties of alkyl- and arylamine radical cations and the corresponding aminyl radicals. *J. Phys. Chem.*, 100:17539–17543, 1996.

Table S7: Coupling scan. “valid”: non-clashing geometries of those computed.

amine	approach	r_{N5N} (Å)	valid	median $ t $ (meV)	max $ t $ (meV)	min contact (Å)
methylamine	face	3.0	5/5	184.6	311.6	2.32
methylamine	face	3.5	5/5	52.8	105.6	2.72
methylamine	face	4.0	5/5	24.6	37.9	3.12
methylamine	face	4.5	5/5	5.1	31.1	3.53
methylamine	face	5.0	5/5	1.9	14.7	3.96
methylamine	face	5.5	5/5	1.0	4.2	4.40
methylamine	face	6.0	5/5	0.3	1.1	4.85
methylamine	edge	3.0	0/3	–	–	0.77
methylamine	edge	3.5	0/3	–	–	0.93
methylamine	edge	4.0	0/3	–	–	1.29
methylamine	edge	4.5	2/3	97.0	170.8	1.71
methylamine	edge	5.0	3/3	23.1	81.0	2.17
methylamine	edge	5.5	3/3	8.6	35.0	2.65
methylamine	edge	6.0	3/3	3.2	12.7	3.13
benzylamine	face	3.0	0/5	–	–	0.61
benzylamine	face	3.5	2/5	47.1	52.3	0.89
benzylamine	face	4.0	2/5	5.2	7.9	1.31
benzylamine	face	4.5	3/5	6.1	35.2	1.77
benzylamine	face	5.0	5/5	4.2	21.1	2.25
benzylamine	face	5.5	5/5	1.0	8.3	2.73
benzylamine	face	6.0	5/5	0.9	2.0	3.22
benzylamine	edge	3.0	0/3	–	–	1.00
benzylamine	edge	3.5	0/3	–	–	1.00
benzylamine	edge	4.0	0/3	–	–	0.92
benzylamine	edge	4.5	0/3	–	–	0.82
benzylamine	edge	5.0	0/3	–	–	1.00
benzylamine	edge	5.5	0/3	–	–	1.36
benzylamine	edge	6.0	1/3	0.4	0.4	1.78

Table S8: One-electron potentials in water.

couple	DFT (V vs NHE)	experiment	error (V)
$\text{Me}_2\text{NH}^{\bullet+}/\text{Me}_2\text{NH}$	+0.73	+1.300	-0.57
$\text{MeNH}_2^{\bullet+}/\text{MeNH}_2$	+1.26	–	–
$\text{BnNH}_2^{\bullet+}/\text{BnNH}_2$	+1.45	–	–
$\text{Fl}/\text{Fl}^{\bullet-}$ (exp.: FMN, pH 7)	-1.16	-0.313	-0.84

- [3] S. G. Mayhew. The effects of pH and semiquinone formation on the oxidation–reduction potentials of flavin mononucleotide. A reappraisal. *Eur. J. Biochem.*, 265:698–702, 1999.
- [4] O. Efimova and P. J. Hore. Role of exchange and dipolar interactions in the radical pair model of the avian magnetic compass. *Biophys. J.*, 94:1565–1574, 2008.

Table S9: Raw DFT driving force of the contact pair.

continuum	amine	separated (eV)	$r = 4 \text{ \AA}$	$5 \text{ \AA}$	$6 \text{ \AA}$
$\varepsilon = 4$	methylaniline	3.55	2.65	2.83	2.95
$\varepsilon = 4$	benzylaniline	3.59	2.69	2.87	2.99
$\varepsilon = 10$	methylaniline	2.83	2.47	2.55	2.59
$\varepsilon = 10$	benzylaniline	2.97	2.61	2.69	2.73

Table S10: Cryptochrome response (%).

J_0 (μT)	J (MHz)	powder (%)	$\perp$ (%)	$\parallel$ (%)
8.00e12	-1.26	+0.417	+0.494	+0.027
8.00e12	+1.26	+0.386	+0.464	-0.023
1.01e13	-1.58	+0.395	+0.469	+0.015
1.01e13	+1.58	+0.384	+0.457	-0.030
1.27e13	-1.99	+0.371	+0.454	-0.005
1.27e13	+1.99	+0.392	+0.461	-0.034
1.60e13	-2.51	+0.403	+0.498	+0.005
1.60e13	+2.51	+0.413	+0.491	-0.029
2.01e13	-3.16	+0.470	+0.561	+0.021
2.01e13	+3.16	+0.435	+0.536	-0.020
2.53e13	-3.98	+0.573	+0.681	+0.027
2.53e13	+3.98	+0.444	+0.580	-0.012
3.18e13	-5.01	+0.634	+0.751	+0.042
3.18e13	+5.01	+0.404	+0.483	+0.007
4.01e13	-6.30	+0.559	+0.622	+0.034
4.01e13	+6.30	+0.334	+0.407	+0.025
5.05e13	-7.93	+0.383	+0.420	+0.061
5.05e13	+7.93	+0.242	+0.320	-0.005
6.35e13	-9.99	+0.441	+0.534	+0.033
6.35e13	+9.99	+0.242	+0.299	+0.049
8.00e13	-12.57	+0.492	+0.589	+0.015
8.00e13	+12.57	+0.168	+0.234	+0.021
1.01e14	-15.83	+0.282	+0.372	+0.024
1.01e14	+15.83	+0.153	+0.174	+0.081
1.27e14	-19.93	+0.085	+0.115	-0.060
1.27e14	+19.93	+0.150	+0.131	+0.126
1.60e14	-25.09	+0.061	+0.078	+0.020
1.60e14	+25.09	+0.185	+0.119	+0.214
2.01e14	-31.58	+0.044	+0.049	+0.023
2.01e14	+31.58	+0.055	+0.078	+0.063
2.53e14	-39.76	-0.009	-0.004	-0.013
2.53e14	+39.76	+0.036	+0.041	+0.019
3.18e14	-50.05	+0.015	+0.017	+0.009
3.18e14	+50.05	+0.068	+0.063	+0.041
4.01e14	-63.01	+0.010	+0.011	+0.007
4.01e14	+63.01	+0.052	+0.045	+0.031
5.05e14	-79.33	+0.004	+0.005	+0.003
5.05e14	+79.33	+0.039	+0.049	+0.015
6.35e14	-99.87	+0.003	+0.003	+0.002
6.35e14	+99.87	+0.034	+0.036	+0.019
8.00e14	-125.73	+0.001	+0.002	+0.001
8.00e14	+125.73	+0.011	+0.017	+0.002